Docker Introduction & Usage

1. Docker Introduction

Docker is an open-source platform and tool that allows you to package, distribute, and run applications within containers. A container is a lightweight, standalone, and executable package that includes everything needed to run an application: code, runtime, system tools, libraries, and configuration files. By utilizing Linux's containerization technology, Docker provides efficient isolation between applications, allowing multiple containers to operate independently on the same physical device.

In essence, Docker simplifies application deployment and improves portability across different environments, making it highly convenient for software development and distribution.

Docker Logo

For more information on Docker, please refer to the tutorials in **“7. Docker Basic Lesson”** or visit the following resources:

Docker Official Website: http://www.docker.com

Docker Chinese Website: https://www.docker-cn.com

Docker Hub (Repository) Official Website: https://hub.docker.com

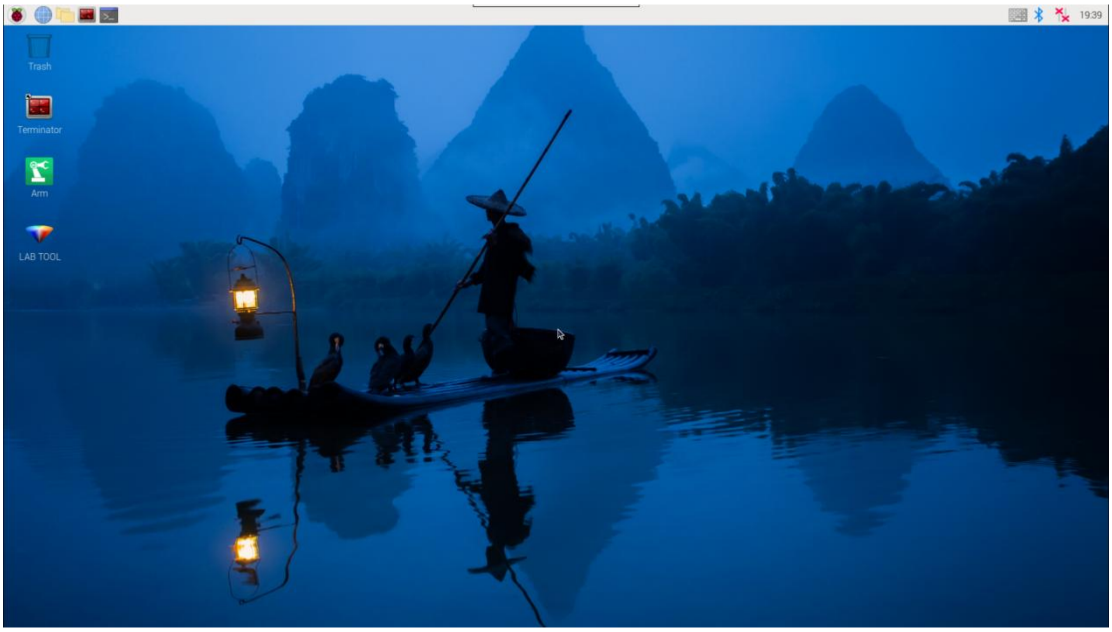
2. Docker Commands

Note: All necessary Docker configurations have been pre-installed in the

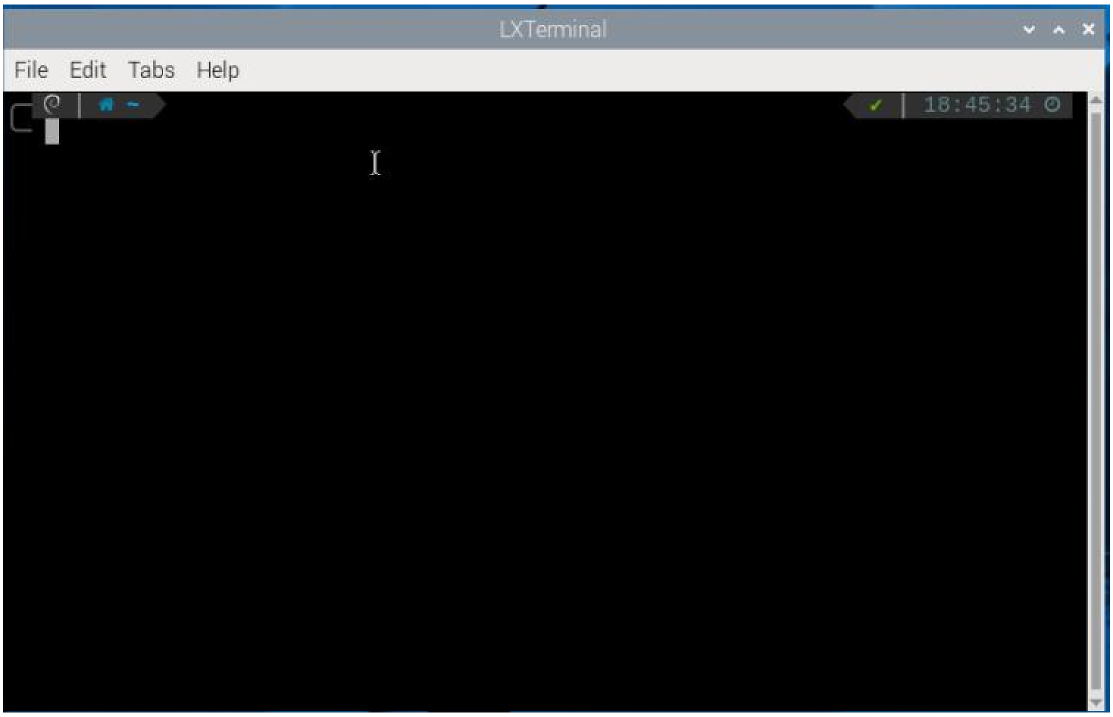
container before shipment.

All of the robot's programs and functional modules run inside a Docker container. The following section introduces some commonly used Docker commands to help you better operate the system.

1) Turn on the device. Follow the instructions in “8. Remote Tool Installation and Container Access/ 8.1 VNC Installation and Connection" to connect to the computer via VNC.



2) Click  in the upper left corner of the system desktop to open the terminal window.



2.1 Check Container

Command parameters: docker ps [OPTIONS]

Commonly used parameters:

① -a: Lists all containers, including those that are currently running and those that have exited

② -l: Displays the most recently created container

③ -n=?: Shows the n most recently created containers

④ -q: Quiet mode—displays only the container IDs

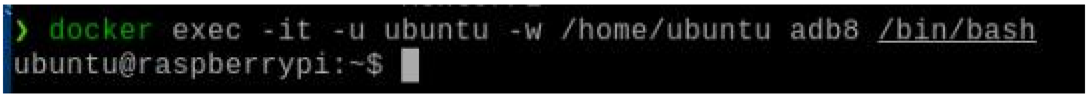
Enter the following command and press “Enter” to display a list of running and previously run containers. The output will show the “container ID,” which is the unique identifier for each container, the “image,” which indicates the name of the image used, the “created” time, which shows when the container was created, and the “status,” which indicates the current state of the container.

```
docker ps -a
```

2.2 Enter Container

Obtain the unique container ID from the output of "2.1 Check Container." Then, enter the following command to access the container containing the function programs.

docker exec -it -u ubuntu -w /home/ubuntu adb8 /bin/bash（The container ID can be abbreviated as long as it remains unique to the container.）

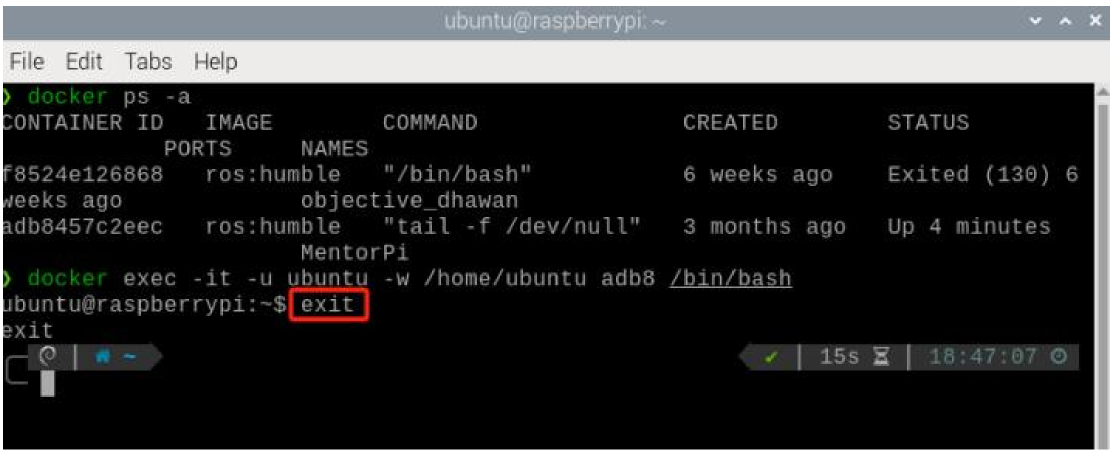


2.3 Exit Container

There are two commands for exiting containers:

Method 1: Enter the command below and press “Enter”. This stops the container and exits it.

exit



Method 2: Use “crtl+P+Q”. This detaches from the container without stopping it.

Then use docker ps to verify that it is still running:

```
docker  ps
```

```
❯ docker ps
CONTAINER ID   IMAGE         COMMAND               CREATED         STATUS
PORTS     NAMES
adb8457c2eec   ros:humble    "tail -f /dev/null"   3 months ago    Up 5 minutes
          MentorPi
                                                                  ✓ │ 18:47:44
```